

UNITED NATIONS SDG IMPACT REPORT

Technological Contribution to Global Cyber Resilience, Economic Security & Innovation

TEAM NAME

TEAM SPARK

INSTITUTION

Sreenidhi Institute of Science and Technology

PROJECT NAME

VibeGuard v3.0

UN FRAMEWORK

2030 Agenda for SDGs

TEAM MEMBERS:

B. Poojitha 24311A6201

Y. Bhavya Sri 24311A6203

A. Ranaveer 24311A6211

M. Gagan Sai 24311A6212

United Nations 2030 Agenda Context: In an increasingly interconnected global economy, software represents the foundational infrastructure of financial systems, healthcare networks, public utilities, and educational institutions. **VibeGuard v3.0** actively advances the UN Sustainable Development Goals by safeguarding digital supply chains, fortifying institutional cyber infrastructure, promoting sustainable economic productivity, and reducing developer computational waste.

01 VibeGuard UN SDG Alignment Matrix

SDG GOAL	OFFICIAL UN GOAL TITLE	IMPACT TYPE	VIBEGUARD DIRECT CONTRIBUTION & IMPACT METRIC
SDG 9	Industry, Innovation & Infrastructure	DIRECT / PRIMARY	Builds resilient digital infrastructure by safeguarding source code and software supply chains from injection attacks and zero-day vulnerabilities.
SDG 16	Peace, Justice & Strong Institutions	DIRECT / PRIMARY	Strengthens institutional governance by preventing data theft, unauthorized credential exposure, and corporate espionage via autonomous security gates.
SDG 8	Decent Work & Economic Growth	SUPPORTIVE	Enhances engineering productivity and preserves digital economic value by eradicating alert fatigue and stopping breach-induced financial damages.
SDG 12	Responsible Consumption & Production	SUPPORTIVE	Reduces high-carbon cloud CI/CD server compute cycles through high-speed, local client-side compiled Go and Rust execution (<100ms scans).

02 Detailed Contribution to Primary UN SDGs

Target 9.1 & 9.4: Develop quality, reliable, sustainable and resilient infrastructure; upgrade infrastructure and retrofit industries with increased resource-use efficiency.

In modern society, software repositories are as critical as bridges and power grids. Critical public and private systems rely on open-source code libraries. When vulnerabilities (such as Log4j or heartbleed-class CVEs) infiltrate these systems, the infrastructure is crippled.

VibeGuard's Engineering Contribution:

- **Defends Critical Digital Infrastructure:** Intercepts software supply-chain vulnerabilities across Go, PyPI, npm, and Cargo ecosystems prior to repository merging.
- **Real-Time Threat Intelligence Integration:** Connects local workflows directly to the Google Open Source Vulnerabilities (OSV) database, ensuring developer tools possess live zero-day advisory awareness.
- **Technological Innovation:** Pioneered a dual-engine architecture combining Rust (for zero-overhead AST regex evaluation) with Go (for platform orchestration and asynchronous concurrency), establishing a new benchmark in developer security tooling.

GOAL 16: PEACE, JUSTICE & STRONG INSTITUTIONS

16

Promote peaceful and inclusive societies for sustainable development, provide access to justice and build accountable institutions

Target 16.4 & 16.6: Significantly reduce illicit financial and arms flows, strengthen the recovery and return of stolen assets and combat all forms of organized crime; develop effective, accountable and transparent institutions.

Cyber warfare, state-sponsored cyber espionage, and ransomware cartels exploit exposed API credentials, database connection strings, and insecure transport layers to infiltrate government bodies, healthcare systems, and universities.

VibeGuard's Engineering Contribution:

- **Autonomous Pre-Push Firewall:** Implements a non-bypassable, fail-closed barrier at git push time that strictly halts code containing exposed AWS tokens, GitHub private keys, Slack credentials, or database passwords.
- **Safeguarding Institutional Data Sovereignty:** Prevents unauthorized exfiltration and weaponization of corporate and civic information assets before public commit propagation.
- **100% Client-Side Privacy Guarantee:** In contrast to cloud-based scanners that mandate sending proprietary intellectual property to third-party servers, VibeGuard performs all source code analysis locally, ensuring complete privacy for sensitive institutional data.

03 Contribution to Supportive UN SDGs

GOAL 8: DECENT WORK & ECONOMIC GROWTH

08

Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all

Target 8.2: Achieve higher levels of economic productivity through diversification, technological upgrading and innovation.

- **Eradicating Developer Alert Fatigue:** Traditional static analysis tools inundate engineers with hundreds of false alarms, destroying software development velocity. VibeGuard's contextual filtering and grouped reports restore productive developer focus.
- **Mitigating Economic Devastation from Data Breaches:** According to IBM's Cost of a Data Breach Report, the average corporate breach costs \$4.45M. VibeGuard prevents breaches at zero software cost, protecting economic enterprise vitality.

GOAL 12: RESPONSIBLE CONSUMPTION & PRODUCTION

12

Ensure sustainable consumption and production patterns

Target 12.2: Achieve the sustainable management and efficient use of natural resources.

- **Green Software Engineering Principles:** Running heavy containerized scanners on cloud CI/CD servers for every Git commit consumes substantial electrical energy. VibeGuard's compiled binary executes in <100ms on the developer's local machine, reducing data center server load and overall carbon emissions.

Summary: By shifting security from reactive post-deployment forensics to proactive pre-push prevention, **TEAM SPARK's VibeGuard v3.0** translates the UN Sustainable Development Goals into tangible, high-impact cyber engineering practice.